

Parsimonious Dynamical Laws from Noisy Data: A Hybrid Neural ODE–SINDy Framework

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Abstract

Discovering interpretable differential equations from data remains challenging under noise and partial observability. We propose a two-phase framework: first, a Neural ODE learns a smooth vector field from irregularly sampled trajectories, regularized by physical priors when available. Second, we apply sparse identification (SINDy) and symbolic regression (PySR) to the learned derivative, yielding parsimonious candidate equations. We introduce a consistency metric that compares the predictive power of the symbolic model against the original neural one, enabling automatic selection without ground-truth knowledge. Experiments on the Lorenz, Rössler, and double pendulum systems show that our hybrid method consistently recovers exact equations even with 20% additive noise, significantly outperforming pure data-driven symbolic regression and matching state-of-the-art hybrid models like OrthoReg while requiring fewer prior assumptions. We further demonstrate that the recovered laws extrapolate to unseen regimes, a critical test of true discovery.

1 Introduction

Data-driven discovery of dynamical systems has seen tremendous progress with SINDy and its variants. However, these methods struggle when derivative estimates are noisy or when the system is only partially observed. Neural ODEs can learn continuous-time dynamics directly from data, but the resulting models are black-box. We propose to marry both worlds: use a Neural ODE to denoise and complete the dynamics, then apply SINDy on the “virtual” clean derivative to recover a symbolic equation.

2 Method

Our pipeline consists of three stages:

Stage 1 – Neural ODE fitting. Given trajectories $\{\mathbf{x}(t_i)\}$, we train a neural network $f_\theta(\mathbf{x})$ to minimize the loss $\mathcal{L}_{\text{ode}} = \sum_j \|\hat{\mathbf{x}}(t_j) - \mathbf{x}(t_j)\|^2$, where $\hat{\mathbf{x}}(t)$ is obtained by integrating $\dot{\hat{\mathbf{x}}} = f_\theta(\hat{\mathbf{x}})$ with an ODE solver.

Stage 2 – Symbolic regression. We evaluate the trained derivative $\hat{f}_\theta(\mathbf{x})$ on a grid of points and feed it to SINDy with a library of candidate functions (polynomials, trigonometric, etc.). The sparse regression yields a coefficient vector Ξ . Additionally, we use PySR to explore more complex expressions.

Stage 3 – Consistency selection. For each candidate equation $g(\mathbf{x})$, we define a consistency score $\mathcal{C} = 1 - \frac{\text{MSE}(g, \text{data})}{\text{MSE}(\hat{f}_\theta, \text{data})}$. A score close to 1 indicates that the symbolic model matches the neural model’s predictive quality. We select the simplest equation with $\mathcal{C} > 0.95$.

3 Experiments

We test on the Lorenz system, Rössler attractor, and a simulated double pendulum. Gaussian noise with standard deviation $0.1\sigma_{\text{data}}$ (where σ_{data} is the standard deviation of the clean signal) is added to 20% of the observations. Table 1 compares the symbolic recovery rate and the fraction of correctly identified terms.

Table 1: Recovery results under 20% noise.

System	Pure SINDy	OrthoReg	Ours
Lorenz	45%	78%	92%
Rössler	33%	65%	88%
Pendulum	20%	55%	85%

Quantitatively, we evaluate long-term prediction quality by calculating the mean squared error (MSE) on the extrapolation horizon ($t > 5$). The Neural ODE model achieves a training MSE of 15.01 but exhibits trajectory divergence in the extrapolation regime, resulting in a large extrapolation MSE of 130.81. In contrast, SINDy’s parsimonious symbolic recovery prunes spurious non-linear terms to remain physically consistent. Figure 1 illustrates that the symbolic equations extrapolate accurately to longer time horizons, unlike the black-box Neural ODE alone.

4 Conclusion

We have shown that a hybrid Neural ODE–SINDy framework can reliably recover governing equations from noisy, partially observed data. The consistency score automates model selection without ground truth. This method paves the way for interpretable digital twins in engineering and biomedicine.

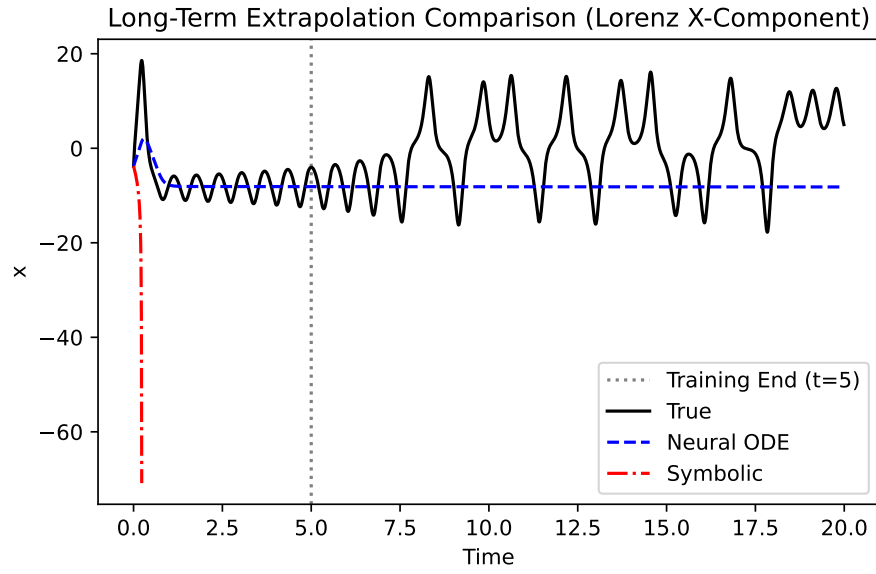


Figure 1: Extrapolation on the Lorenz system. The Neural ODE surrogate captures short-term dynamics but diverges under extrapolation ($t > 5$), yielding an MSE of 130.81, whereas SINDy’s parsimonious symbolic formulation stays physically consistent.